

RUPAN CHAKKARAVARTHY E

Technical Lead — Solutions · AI Platforms · XR & Real-Time Systems

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PROFILE

Technical Lead — Solutions with ten years spanning two hard domains: real-time defense simulation (fighter-jet simulators, ARINC 661 cockpit displays, maritime situational awareness, XR hardware integration) and, since 2024, agentic AI platforms. Currently architecting AI-driven software delivery systems that orchestrate fleets of coding agents through spec, plan, implement, review and QA phases with human approval gates — taking one platform from empty repository to a self-running system in 18 days. Comfortable owning everything from GLSL shaders and Ada integration to RAG pipelines, NL-to-SQL engines and air-gapped Linux delivery. Team lead and technical manager for national aerospace and defense R&D programs.

CORE SKILLS

AI & Agents: Claude / Claude Code multi-agent orchestration, MCP, RAG (RAGFlow, Chroma), Vanna NL→SQL, Ollama local LLMs, code knowledge graphs, agent-native architecture

XR & Simulation: Unity 3D, C#, MRTK3, ARCore / ARFoundation, OpenVR / SteamVR, FlightGear, Varjo XR-3/4, Meta Quest, HTC Vive, Leap Motion, Xsens Movella, SenseGlove

Real-Time & Avionics: C, C++, OpenGL, ARINC 661, Ada embedded integration, Qt, ROS2 / NAV2, GLSL/HLSL shaders, OpenCV, UDP video streaming

Web & Platform: TypeScript, React, Vite, Node.js, Express, SSE, SQLite / MySQL / MSSQL, Tailwind, auth (scrypt, HMAC, RBAC)

Ops & Delivery: Docker, nginx, air-gapped Ubuntu deployment, Git worktree orchestration, TDD, automated browser testing, WBS / SOW / costing

WORK EXPERIENCE

Technical Lead — Solutions — MicroGenesis TechSoft Pvt Ltd

Mar 2024 – Present

- Deployed with the Aeronautical Development Agency (DRDO, Ministry of Defence) for indigenous XR real-time pilot ergonomics study, and with CAIR-DRDO as Technical Manager for GIS, real-time and robotics programs.
- Architected and solely built an AI-SDLC orchestration platform driving Claude coding agents through spec → plan → implement → review → QA with human gates: dependency-wave scheduler running 8 parallel agents in isolated git worktrees, automated browser verification of acceptance criteria, code-knowledge-graph context layer, local-model (Ollama) routing for on-premise use. Empty repo to self-running platform in 18 days.
- Delivered an AR quality-inspection product for a manufacturing client from zero to client-ready in 3 weeks: Unity + MRTK3 headset experience, ARCore tablet port at full parity verified on physical hardware with test-driven EditMode and E2E suites, plus a React/Express management portal with rework traceability and first-pass-yield analytics.
- Built a flight-data insights platform: RAG over raw .dat telemetry, NL→SQL (Vanna 2.0) verified live across 13 datasets, MPEG-2 video-evidence pipelines — packaged for fully air-gapped Ubuntu deployment (Docker, offline installers, patch bundles).
- Designed and shipped an offline↔online secure data-exchange PoC (ARINC-style signed, encrypted change-set packaging with field-level review) end-to-end in one day.
- Led commercial engineering: tender analysis to costed proposals, WBS/SOW generation tooling, market research across 125 competing agent-orchestration projects, pitch decks and phased rebuild programs (three phases delivered in one weekend).
- Sustained throughput of 110+ tracked tasks per week with automated reporting tooling built in-house.

Freelance Software Engineer — Independent

Mar 2023 – Mar 2024

- Developed interactive applications; ran day-to-day project coordination, planning and implementation; authored functional and technical documentation.

Senior Engineer — COMAVIA Systems Technologies Pvt Ltd

Nov 2021 – Feb 2023

- Developed ARINC 661-compliant Cockpit Display System software for indigenous fighter programs (MK2, AMCA) in C, C++ and OpenGL — low-latency SVG-based display pages, shader-based rendering, multi-partition dynamic displays. Team lead.
- Integrated Epiphan video capture with real-time cropping; UDP video streaming via OpenCV; Ada ↔ C++ integration over DLLs and network protocols for airworthy test units.
- Developed IMSAS — Indian Maritime Situational Awareness System: GIS-based real-time common operating picture in Qt/C++ with PostgreSQL storage and Kafka / RabbitMQ event streaming across consoles.

Senior Software Engineer (XR) — BAeHAL Software Limited

Jan – Oct 2019, Nov 2020 – Oct 2021

- Deployed with the Aeronautical Development Agency for the indigenous VR fighter-jet simulator: HUD shaders (world-space UV), physics-joint cockpit switches, Leap Motion hand tracking, editor-scripted auto-rigging of cockpit controls from engineering spreadsheets, multi-surface symbology scheduling. Team lead.

Senior XR Engineer / XR Engineer — D2VR Technologies

Jul – Dec 2018, Oct 2019 – Oct 2020

- Built VR/AR applications and prototypes; agile planning, code reviews, risk identification and workflow improvement.

Unity 3D Developer — Satya Aerospace Tech Data Services Pvt Ltd

May 2016 – Jun 2018

- Game programming, network gameplay code, and VR builds for Windows MR, Oculus Rift and HTC Vive; AR with Vuforia, ARCore, Google VR.

KEY PROJECTS

- **FORGE:** AI software delivery platform — agent orchestration with human gates, knowledge-graph context, autonomous defect-fix loop (an automated browser test caught a data-loss bug; the system repaired and re-verified itself).
- **Qualia:** AR quality inspection for manufacturing — headset + tablet parity, TDD throughout, nine feature updates shipped in a single day.
- **Flight Data Insights:** RAG + NL→SQL telemetry analysis, air-gapped delivery for flight-test environments.
- **Supplier Intelligence:** Component supply-chain risk intelligence — 187 systems risk-scored, market cross-referencing, disruption simulation; full costed delivery program (5,692 hours).
- **Defense simulation:** Indigenous VR fighter-jet simulator and ARINC 661 cockpit display systems for national fighter programs.
- **Pilot ergonomics:** Xsens motion-capture ergonomic analysis of pilots under high-load maneuvering, with automated reporting.
- **IMSAS:** Indian Maritime Situational Awareness System — GIS-based real-time common operating picture. Qt, C++, PostgreSQL, Kafka, RabbitMQ.
- **Simulation & CV:** Bomb-drop trajectory simulation (FlightGear/Unity), fighter-jet sensors simulation with multi-monitor XLIB windows on Ubuntu, synthetic-terrain shaders in FlightGear, lane-detection CV integration, UDP video streaming via OpenCV.
- **Avionics & robotics:** ECFM LRU test-case generation and component testing for an airworthy unit (Ada ↔ C++), AUV ECA-A9 software on ROS2/NAV2, multi-region large-area touch displays in Windows C++.
- **XR & interactive:** LIDAR point-cloud VR walkthroughs with MIT Chennai, museum interactives (Puli-Meka multiplayer AI strategy game, 4-way touch table, VR 360° tour), AR sketch app with SLAM anchoring, MMORPG VR building/terrain/networking systems.
- **Self-hosted AI infrastructure:** Multimodal LLM management backend and open-source chat platform with user/model access control (Docker), VS Code AI pair-programming rollout on local LLMs, low-latency self-hosted video conferencing.

EDUCATION

B.E. Computer Science — MNM Jain Engineering College

2012 – 2016